

ANTSIRANANA, Madagascar

WMO: 670090

Lat: 12.349S

Lon: 49.292E

Elev: 114

StdP: 99.96

Time Zone: 3.00 (E03)

Period: 90-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.0	19.8	12.3	9.0	24.9	13.2	9.6	24.7	13.7	25.8	12.9	25.7	4.0	160	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	7.9	33.0	24.7	32.2	24.7	31.9	24.7	27.2	30.7	26.7	30.3	26.2	29.8	5.4	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
26.2	21.9	29.7	25.7	21.3	29.3	25.2	20.6	28.6	87.0	30.8	84.6	30.4	82.4	30.0	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.5	11.9	11.1	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	26.3	26.9	26.9	27.0	27.0	26.5	25.3	24.6	24.8	25.2	26.4	27.5	27.8
	DBStd	1.36	1.10	0.88	0.82	0.84	0.82	0.98	0.84	0.90	0.91	0.77	0.94	0.91
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	5959	522	472	528	511	512	458	454	460	457	507	526	552
	CDD18.3	2918	264	239	269	261	253	208	196	202	207	249	276	294
	CDH23.3	24025	2284	2081	2412	2346	2045	1397	1222	1272	1402	2002	2663	2901
	CDH26.7	7811	670	604	777	812	668	401	309	329	418	711	1032	1078
Wind	WSAvg	5.5	3.1	2.5	3.1	4.3	6.3	6.4	7.1	8.0	8.3	6.8	5.9	3.8
Precipitation	PrecAvg	1135	331	303	171	60	15	20	21	20	9	13	49	164
	PrecMax	1711	850	672	476	180	89	73	75	54	34	57	193	493
	PrecMin	624	84	36	27	1	1	1	0	0	0	0	2	12
	PrecStd	332	197	139	94	48	16	19	18	13	9	13	49	128
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.1	32.1	32.7	33.0	32.0	31.2	30.9	30.9	31.2	32.3	33.8	33.9
		MCWB	25.3	26.2	26.5	25.2	24.1	22.6	20.9	21.3	21.9	23.5	25.3	24.4
	2%	DB	31.7	31.2	31.9	32.1	31.2	30.4	29.9	30.0	30.7	31.8	32.8	33.0
		MCWB	25.5	25.9	25.9	25.0	23.8	22.0	20.8	21.2	21.8	23.2	24.4	24.5
	5%	DB	30.9	30.8	31.2	31.5	30.7	29.9	29.1	29.2	30.0	31.0	32.0	32.1
		MCWB	25.5	25.8	25.7	24.9	23.6	21.6	20.5	20.8	21.8	22.7	24.2	24.6
	10%	DB	29.9	30.0	30.5	30.8	30.0	29.1	28.6	28.8	29.2	30.2	31.2	31.2
		MCWB	25.3	25.7	25.6	24.7	23.4	21.2	20.5	20.6	21.5	22.3	23.9	24.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.4	27.7	27.7	27.2	26.2	24.1	23.6	23.5	24.8	25.5	27.6	27.6
		MCDB	30.4	30.5	31.2	31.4	30.3	29.9	27.7	28.7	29.4	30.7	31.7	30.8
	2%	WB	26.6	26.9	26.9	26.2	25.3	23.2	22.4	22.5	23.7	24.5	26.6	26.6
		MCDB	29.8	30.0	30.7	30.4	29.6	28.4	27.4	27.9	28.7	29.7	30.8	30.3
	5%	WB	26.1	26.4	26.3	25.7	24.5	22.6	21.7	21.8	22.9	23.8	25.8	26.0
		MCDB	29.3	29.5	30.0	29.9	29.1	27.7	26.9	27.2	28.1	29.1	29.8	30.0
	10%	WB	25.6	25.9	25.9	25.2	23.9	22.1	21.2	21.3	22.3	23.3	25.0	25.5
		MCDB	28.8	29.0	29.5	29.5	28.8	27.3	26.5	26.7	27.4	28.4	29.0	29.4
Mean Daily Temperature Range	5% DB	MDBR	6.7	6.7	7.2	7.6	7.8	8.5	8.7	8.6	8.6	8.6	8.3	7.9
		MCDBR	8.1	7.9	8.3	8.8	8.6	9.1	9.4	9.2	9.4	9.4	9.2	8.9
		MCWBR	4.0	4.2	4.4	4.0	4.0	3.9	4.0	3.7	4.1	3.9	3.8	3.6
	5% WB	MCDBR	7.5	7.3	7.5	8.0	8.3	8.5	8.5	8.7	8.5	8.7	8.6	8.0
		MCWBR	4.2	4.4	4.6	4.3	4.4	4.2	4.2	4.2	4.7	4.3	4.6	4.2
Clear-Sky Solar Irradiance	taub		0.414	0.419	0.400	0.382	0.371	0.363	0.357	0.363	0.374	0.388	0.401	0.404
	taud		2.509	2.492	2.553	2.592	2.597	2.585	2.594	2.564	2.516	2.487	2.481	2.514
	Ebn at Noon		930	919	918	904	883	874	889	910	929	937	935	938
	Edh at Noon		114	116	106	97	91	90	90	98	108	115	117	113
All-Sky Solar Radiation	RadAvg		5.63	5.59	5.69	5.67	5.15	4.75	4.95	5.52	6.31	6.79	6.71	6.31
	RadStd		0.42	0.47	0.37	0.22	0.18	0.12	0.11	0.12	0.17	0.18	0.25	0.46

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page